Knowledge Distillation & Temperature.

A small student network learns to imitate a large teacher's *soft* probability distribution, not just its hard answer. We derive the temperature-softened softmax, the three-term distillation loss, and the T^2 correction — and compute every number by hand, nothing summarised or deferred.

THE EQUATION

$$\mathcal{L} \;=\; \alpha\,\mathcal{L}_{\mathrm{CE}} \;+\; \beta\,T^2\,\mathrm{KL}\big(p^{\mathcal{T}}(T)\,\|\,p^{\mathcal{S}}(T)\big) \;+\; \gamma\,\big(1-\cos(h^{\mathcal{S}},h^{\mathcal{T}})\big)$$

Worked example. Teacher logits $[3.0, 1.0, 0.5, -1.0]$, student logits $[2.0, 1.2, 0.3, -0.5]$ over 4 classes, softened at $T = 1$ and $T = 3$.

Topic 1 of 2 in this module · companion to the course page · v1.0

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Contents

1 The claim: learn from soft targets

A trained *teacher* (BERT-base, 110M params) produces, for each input, a probability distribution over classes. The hard label keeps only the argmax; the full distribution carries far more. *Knowledge distillation* trains a smaller *student* to reproduce that distribution. The total objective combines three terms:

$$\boxed{\mathcal{L} = \alpha\,\mathcal{L}_{\mathrm{CE}}(\text{student}, y) \;+\; \beta\,\mathcal{L}_{\mathrm{KD}} \;+\; \gamma\,\mathcal{L}_{\cos}} \tag{1}$$

Symbol	Shape	Meaning
$z^{\mathcal{T}}$, $z^{\mathcal{S}}$	$\mathbb{R}^C$	teacher / student logits over C classes
T	scalar	temperature; $T > 1$ softens, $T = 1$ recovers ordinary softmax
$p^{\mathcal{T}}(T)$, $p^{\mathcal{S}}(T)$	$\mathbb{R}^C$	softened distributions, eq. (2)
$h^{\mathcal{S}}$, $h^{\mathcal{T}}$	$\mathbb{R}^H$	hidden states to be aligned
α, β, γ	scalars	loss weights ($\alpha+\beta+\gamma$ need not be 1)

The hard label says "this is a dog." The teacher's soft distribution says "this is most likely a dog, but it looks a bit like a wolf and nothing at all like a teapot." Those relative magnitudes — the **dark knowledge** — tell the student about inter-class geometry that a one-hot target erases entirely. Matching them is a far richer supervisory signal than a single index.

2 Temperature-softened softmax

For temperature T, the softened probability of class i is

$$p_i(T) \;=\; \mathrm{softmax}\Big(\tfrac{z}{T}\Big)_i \;=\; \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}. \tag{2}$$

As $T \to \infty$ all $z_i/T \to 0$ and $p \to$ uniform; as $T \to 0^+$ the distribution collapses to one-hot at the argmax. Dividing logits by T shrinks their spread, so larger T *flattens* the distribution and exposes the small-probability tail.

2.1 Worked: teacher distribution at $T = 1$ vs $T = 3$

Teacher logits $z^{\mathcal{T}} = [3.0,\ 1.0,\ 0.5,\ -1.0]$.

At $T = 1$: exponentiate the raw logits.

$$e^{z} = [\,e^{3.0},\ e^{1.0},\ e^{0.5},\ e^{-1.0}\,] = [\,20.0855,\ 2.7183,\ 1.6487,\ 0.3679\,], \quad \textstyle\sum = 24.8204.$$

$$p^{\mathcal{T}}(1) = \tfrac{1}{24.8204}[\,20.0855, 2.7183, 1.6487, 0.3679\,] = [\,0.8092,\ 0.1095,\ 0.0664,\ 0.0148\,].$$

At $T = 3$: scale logits first, $z/3 = [1.0000,\ 0.3333,\ 0.1667,\ -0.3333]$.

$$e^{z/3} = [\,2.7183,\ 1.3956,\ 1.1814,\ 0.7165\,], \qquad \textstyle\sum = 6.0118.$$

$$p^{\mathcal{T}}(3) = \tfrac{1}{6.0118}[\,2.7183, 1.3956, 1.1814, 0.7165\,] = [\,0.4522,\ 0.2321,\ 0.1965,\ 0.1192\,].$$

The peak drops from 0.809 to 0.452 and the tail classes lift from 0.015 to 0.119 — the distribution has flattened, revealing structure the hard label hides.

2.2 Heatmap: how temperature flattens the distribution

	class 0	class 1	class 2	class 3
$p^{\mathcal{T}}(T{=}1)$	**0.809**	0.110	0.066	0.015
$p^{\mathcal{T}}(T{=}3)$	**0.452**	0.232	0.197	0.119

(Cell intensity scales with probability relative to each row's max, capped at 80.)

> Temperature is a magnifying glass on the teacher's tail. At $T = 1$ the student would chase a near one-hot target and learn almost nothing beyond the label. At $T = 3$ the ratios among the non-top classes become loud enough to teach — this is exactly the dark knowledge transfer.

3 The three loss terms

3.1 Distillation KL term and the T^2 correction

The KD term is the KL divergence between teacher and student *softened* distributions, scaled by T^2:

$$\mathcal{L}_{\text{KD}} = T^2\,\text{KL}\big(p^{\mathcal{T}}(T)\,\|\,p^{\mathcal{S}}(T)\big) = T^2 \sum_i p_i^{\mathcal{T}}(T)\,\log\frac{p_i^{\mathcal{T}}(T)}{p_i^{\mathcal{S}}(T)}. \tag{3}$$

Why the T^2? The gradient of the softened cross-entropy w.r.t. a student logit $z_k^{\mathcal{S}}$ is $\frac{1}{T}(p_k^{\mathcal{S}}(T) - p_k^{\mathcal{T}}(T))$. Because the softened probabilities themselves vary on a $\sim 1/T$ scale, the difference $p^{\mathcal{S}} - p^{\mathcal{T}} \sim O(1/T)$, so each gradient component scales as $1/T^2$. Multiplying the loss by T^2 cancels this, keeping the KD gradient magnitude comparable to $\mathcal{L}_{\text{CE}}$ regardless of the chosen T — so the loss weights α, β stay meaningful when you retune T.

3.2 Worked: KL and $\mathcal{L}_{\text{KD}}$ at $T = 3$

Student logits $z^{\mathcal{S}} = [2.0, 1.2, 0.3, -0.5]$ give, by the same procedure as eq. (2) at $T = 3$,

$$p^{\mathcal{S}}(3) = [\,0.3613,\ 0.2767,\ 0.2050,\ 0.1570\,].$$

With $p^{\mathcal{T}}(3) = [0.4522, 0.2321, 0.1965, 0.1192]$, term by term:

$$\sum_i p_i^{\mathcal{T}} \log\frac{p_i^{\mathcal{T}}}{p_i^{\mathcal{S}}} = \underbrace{0.4522\ln\tfrac{0.4522}{0.3613}}_{0.10148} + \underbrace{0.2321\ln\tfrac{0.2321}{0.2767}}_{-0.04086} + \underbrace{0.1965\ln\tfrac{0.1965}{0.2050}}_{-0.00832} + \underbrace{0.1192\ln\tfrac{0.1192}{0.1570}}_{-0.03277}.$$

$$\text{KL}\big(p^{\mathcal{T}}(3)\,\|\,p^{\mathcal{S}}(3)\big) = 0.019527.$$

Apply the correction:

$$\boxed{\mathcal{L}_{\text{KD}} = T^2\,\text{KL} = 9 \times 0.019527 = 0.175746.}$$

(For comparison, at $T = 1$ the KL is 0.120425 and T^2=1, so $\mathcal{L}_{\text{KD}} = 0.120425$.)

3.3 Hard-label CE and cosine alignment $\mathcal{L}_{\cos}$

The student still sees the true label. With ground-truth class 0 and student probabilities at $T = 1$, $p^{\mathcal{S}}(1) = [0.5834, 0.2621, 0.1066, 0.0479]$:

$$\mathcal{L}_{\text{CE}} = -\log p_0^{\mathcal{S}}(1) = -\log 0.5834 = 0.538887.$$

The hidden-state term pulls the student's representation parallel to the teacher's:

$$\mathcal{L}_{\cos} = 1 - \cos(h^{\mathcal{S}}, h^{\mathcal{T}}), \qquad \cos(a,b) = \frac{a^\top b}{\|a\|\,\|b\|}.$$

Take $h^{\mathcal{S}} = [0.8, 0.1, -0.3, 0.5]$, $h^{\mathcal{T}} = [0.6, 0.2, -0.1, 0.7]$:

$$h^{\mathcal{S}} \cdot h^{\mathcal{T}} = 0.88, \qquad \|h^{\mathcal{S}}\| = 0.994987, \qquad \|h^{\mathcal{T}}\| = 0.948683,$$

$$\cos = \tfrac{0.88}{0.994987 \times 0.948683} = 0.932275, \qquad \mathcal{L}_{\cos} = 1 - 0.932275 = 0.067725.$$

3.4 Putting it together

With weights $(\alpha, \beta, \gamma) = (0.25,\ 0.5,\ 0.25)$ and the $T = 3$ KD term:

$$\begin{aligned}\mathcal{L} &= 0.25(0.538887) + 0.5(0.175746) + 0.25(0.067725)\\ &= 0.134722 + 0.087873 + 0.016931 = \boxed{0.239526}.\end{aligned}$$

4 Properties that matter

1. **Gradient invariance to T.** The T^2 factor makes the KD gradient scale $O(1)$ in T, so a single set of weights (α, β, γ) transfers across temperatures.
2. **KL is asymmetric.** We minimise $\mathrm{KL}(p^{\mathcal{T}} \| p^{\mathcal{S}})$ (forward KL), which is mean-seeking: the student is penalised for assigning low probability anywhere the teacher assigns high probability, so it must cover the whole teacher distribution.
3. **Three complementary signals.** CE anchors to truth, KD transfers dark knowledge, cos aligns internal geometry — DistilBERT uses all three.
4. **Empirical payoff.** DistilBERT is $\sim$40% smaller and $\sim$60% faster than BERT-base while retaining $\sim$97% of its GLUE score.

Forgetting the T^2 factor (or, equivalently, scaling the loss by $1/T^2$) is a classic bug: the KD gradient then shrinks like $1/T^2$, so at $T = 3$ it is $\sim 9\times$ too weak relative to $\mathcal{L}_{\mathrm{CE}}$. Training silently ignores the teacher and the student just memorises hard labels. Always apply T^2 to the KD term — and use the *same* T for both teacher and student inside the KL.

5 Where this sits in the track

Connects to	How
Module 1.1 (BERT)	the teacher; its MLM-pretrained logits are the distillation target
Topic 0.1.3 (Softmax)	eq. (2) is softmax with a temperature knob on the logits
Module 1.3 Topic 2	ALBERT shrinks via parameter sharing; distillation shrinks via a smaller student — two orthogonal compression strategies

A Reproducible (NumPy)

Inputs: teacher logits $[3, 1, 0.5, -1]$, student logits $[2, 1.2, 0.3, -0.5]$, hidden states $h^{\mathcal{S}} = [0.8, 0.1, -0.3, 0.5]$, $h^{\mathcal{T}} = [0.6, 0.2, -0.1, 0.7]$, weights $(0.25, 0.5, 0.25)$, true class 0.

```
import numpy as np
def softmax(z): e = np.exp(z - z.max()); return e / e.sum()
tz = np.array([3.0, 1.0, 0.5, -1.0]); sz = np.array([2.0, 1.2, 0.3, -0.5])
```

```
for T in (1, 3):
    pt, ps = softmax(tz / T), softmax(sz / T)
    kl = np.sum(pt * np.log(pt / ps))
    print(T, np.round(pt, 4), "KD =", round(T*T*kl, 6))
# 1 [0.8092 0.1095 0.0664 0.0148] KD = 0.120425
# 3 [0.4522 0.2321 0.1965 0.1192] KD = 0.175746
Lce = -np.log(softmax(sz)[0])                       # 0.538887
hs, ht = np.array([0.8,0.1,-0.3,0.5]), np.array([0.6,0.2,-0.1,0.7])
cos = hs @ ht / (np.linalg.norm(hs) * np.linalg.norm(ht))
Lcos = 1 - cos                                       # 0.067725
L = 0.25*Lce + 0.5*0.175746 + 0.25*Lcos
print(round(L, 6))                                   # 0.239526
```

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